AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *[GEN26-AOH-I023]*

Team Name: *[ACTUATOR ALLIANCE]*

Domain: *[07]*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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Presentation Outline

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Problem Statement

[In many campuses there is a lot of electrical energy because lights , fans , air conditioner and other electrical systems stay on when the room is empty or not being used. Most current ways of tracking systems don't give upto date information. This makes it hard to manage proper leading to operating cost and wasted power.]

Problem Understanding

- [People who run offices, classrooms, labs, hostels and smart homes are the most affected since they feel the burnt of paying higher cost for electricity]
- [The problem arises mainly due to human negligence toward unnecessary usage of electrical appliances.]
- [It affects users and stakeholders through unnecessary increase in the cost of electricity, and increase in the amount of carbon footprint.]

Existing Gaps & Challenges

- [It requires manual checking of whether the device is turned on or off which is not at all convenient or efficient.]
- [The current system used to maintain the current supplied to our devices is neither efficient nor conserving, it leads a lot of electricity wastage.]
- [Main challenge that remains unsolved is not the electricity which is being used but the energy which is being wasted unnecessarily.]

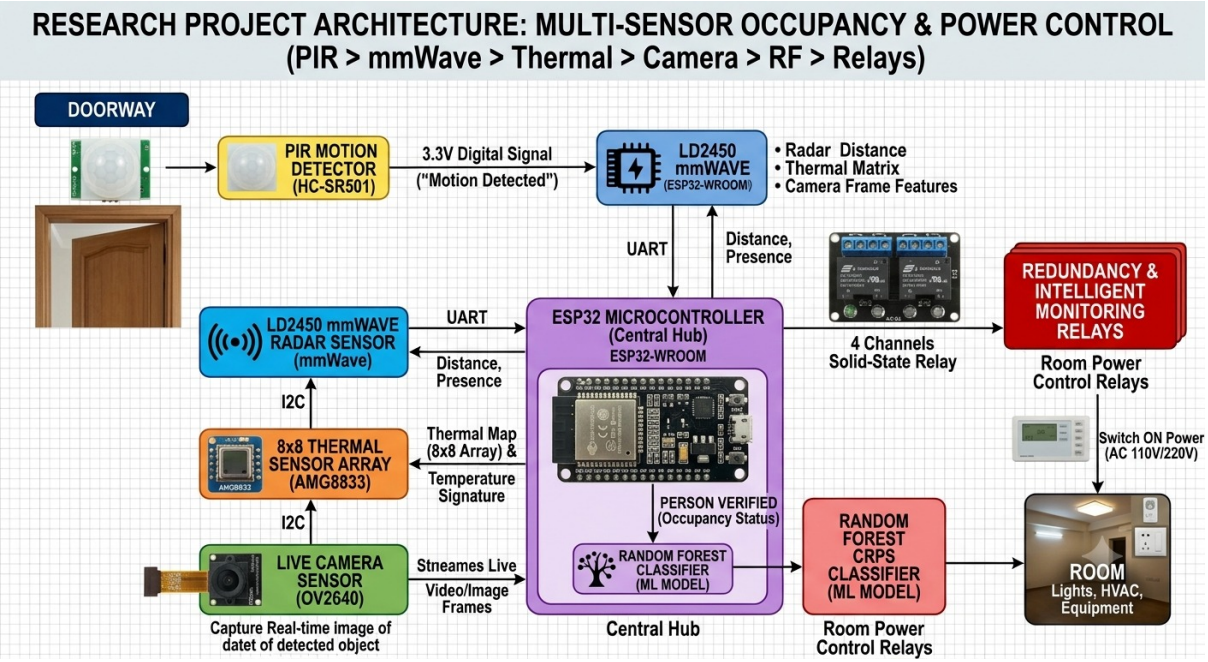
Our Proposed Solution

- [The core idea of our project is to use an IoT based sensor fusion system to detect the presence of humans and turn off the devices where there are no people and reduce the unnecessary usage of electricity.And continuous updation of electricity usage in a specific room through specified dashboard.]

- [The solution works at a high level in the following way:
  1. Our solution uses ESP32-based power detection, MQTT communication, a backend and a React dashboard to monitor and display electrical power availability across campus rooms in near real time.
  2. The system monitors whether electrical supply is present; it does not detect classroom occupancy or automatically switch appliances off. Use an appropriately rated, isolated detection module for mains-voltage monitoring.]

- [Main value delivered to users is the reduced consumption of power and electricity indirectly reducing the electricity costs through automated smart system developed by us.]

Technical Approach



[Explain how the system components work together.]

Tech Stack

Frontend:[C/C]

Backend: [MQPT, Express server,Mongo DB]

AI/ML, APIs, or Hardware: [ESP32 microcontroller,8x8 AMG8833gridEYE, LD2450mmWave Radar,Solid state Relay]

Innovation & Uniqueness

- [Instead of using traditional systems used until now, we use low cost equipment and use sensor fusion to collaborate different sensors to produce high accuracy probability and privacy, making it highly usable and available.]
- [Our approach highly suits the pointed issue because not only does it reduce the wasted electricity and power but it is also readily available at a comparatively low price.]
- [The key differentiator is its low production and availibily cost,its high efficiency and reduced energy consumption.]

Feasibility & Scalability

- **Feasibility:** [Resources and steps needed to build it are:
  1. Resources: ESP32 Core,Sensors Array,Relay Actuation
  2.4-Step Build Process:Circuit Wiring(Connect PIR, radar, thermal, and camera to ESP32 pins.),Data Acquisition(PIR wakes system; radar and thermal capture telemetry.),Model Deployment(Export trained Random Forest Python model to C++ code.),Power Actuation(Verified occupancy toggles intelligent monitoring relays.).]

- **Scalability:** [It can grow or expand through the following areas:
  IoT Mesh Network,Central Dashboard,BMS Integration.]

- **Risks:** [Main constraints and possible mitigations are:
  False Triggers(Ambient heat or sunlight causing false positives.),Complete Privacy(Ensuring zero visual tracking or recording concerns.),High Power Loads(Inrush current damaging control electronics.).]

Cost for Product (Optional)

- Development or hardware cost: [Add estimate]
- Hosting or maintenance cost: [Add estimate]
- Estimated total: [Add amount and currency]

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: [University Lecture Halls:
  Automatically shutting down HVAC and high-power projectors after lectures end and students exit.]
- Use case 2: [Research Labs:
  Powering off non-essential workbench equipment when researchers leave, maintaining zero video privacy risks.]
- Use case 3: [Library Study Pods:
  Detecting stationary occupants studying silently at desks, cutting outlet power only when entirely vacant.]
  Note: Additional slides may be included only for this section.

Impacts and Benefits

- [Benefit to the target users:
  Seamless, hands-free automation without manual switching or camera privacy invasion.]
- [Social, economic, or environmental impact:
  Reduces campus carbon footprint and slashes university utility bills.]
- [Expected measurable outcome:
  25–40 percent reduction in wasted energy during vacant campus hours.]

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG [7]	Target [7.3]	[Affordable and Clean Energy]	[Doubles global energy efficiency by cutting AC appliance power during vacant room windows.]
SDG [13]	Target [13.2]	[Climate]	[Integrates automated climate measures into campus policies to slash carbon emissions.]

Individual Member Contribution

Member	Contribution or responsibility
[Member 1 JISHNU PRABU K]	[COORDINATING THE TEAM AND THE FOUNDER OF SOLUTION]
[Member 2 SHARMILI KRISHNAN]	[RESEARCH ABOUT IoT AND ITS SERVERS]
[Member 3 V GAUTHAM]	[RESEARCH AND DEVELOPMENT]
[Member 4 PEDIREDLA CHARITESH]	[PROJECT DESIGNING]

Yukti ID

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